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(54) **TRANSACTION CARD ASSEMBLY**

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(52) **U.S. Cl.**
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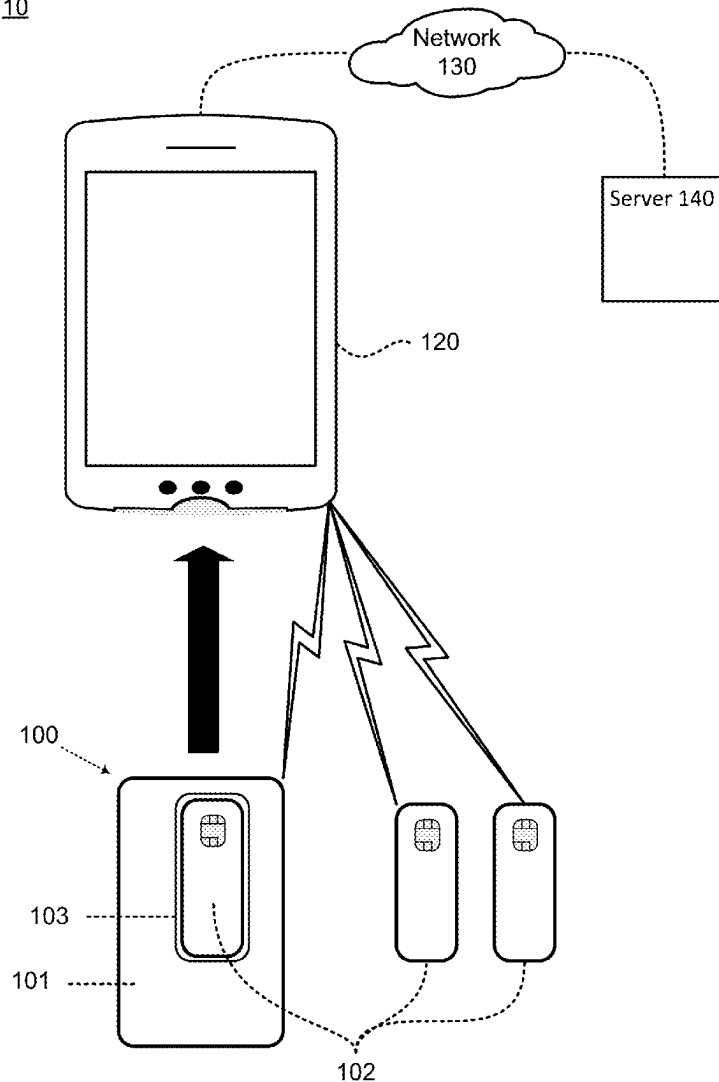
Related U.S. Application Data

(63) Continuation of application No. 18/523,421, filed on Nov. 29, 2023, now Pat. No. 12,307,317, which is a continuation-in-part of application No. 17/539,633, filed on Dec. 1, 2021, now Pat. No. 11,868,833, which is a continuation-in-part of application No. 17/539,636, filed on Dec. 1, 2021, now Pat. No. 11,893,573.

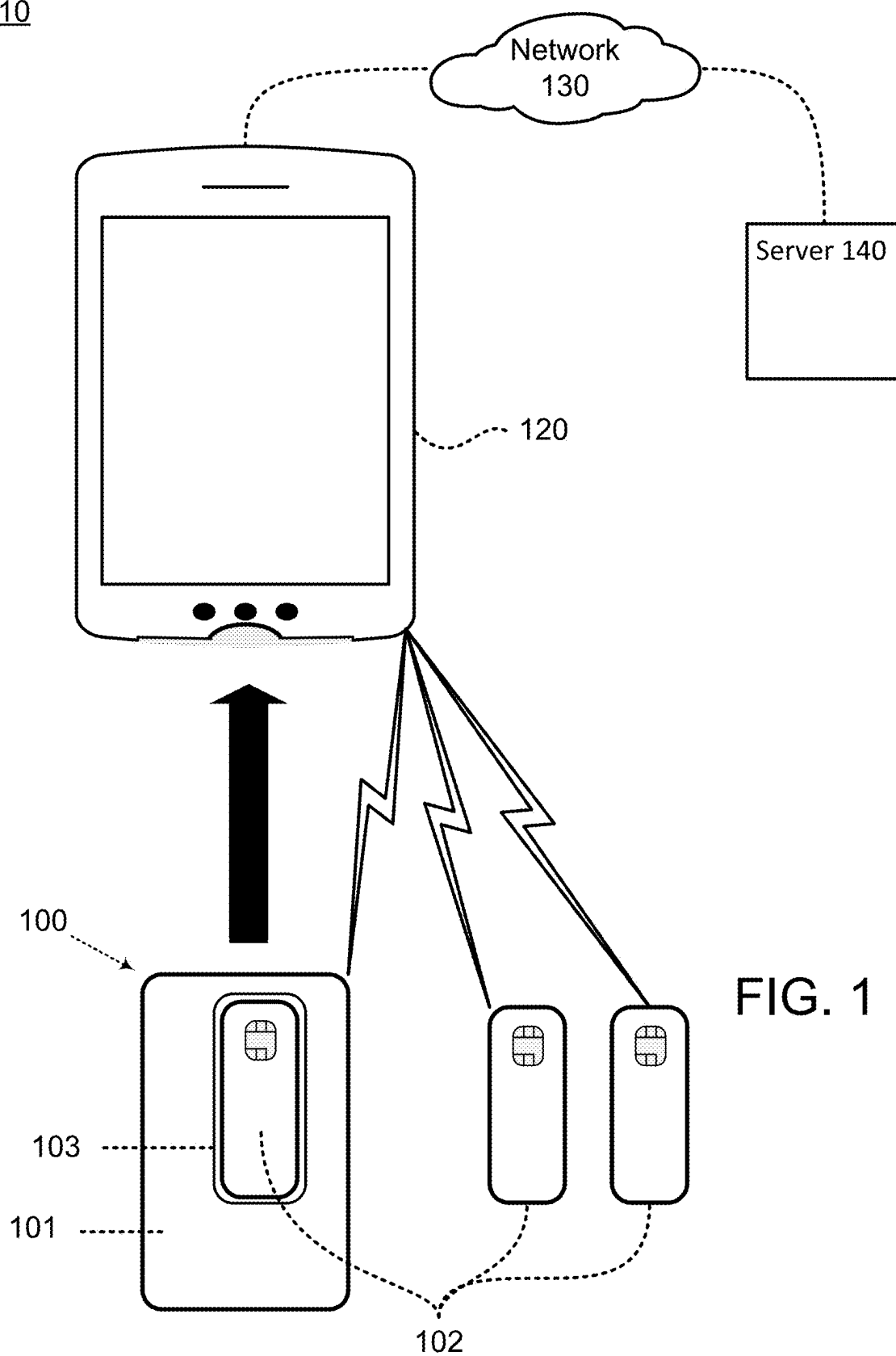
(57) **ABSTRACT**

A modular transaction card assembly includes a card frame having the traditional dimensions of a credit card, and a transaction card that is smaller than a traditional card and that fits into a receptacle of the card frame. Each of the card frame and the transaction card may be capable of performing contactless data transactions individually. The combined assembly of the card frame with the transaction card secured in the receptacle is also capable of performing contact data transactions, and in some instances, with an identifier that is distinct from the card frame or the transaction card alone. The card frame may include a processor enabling it to perform data encryption and authentication of the smaller transaction card.

10



10



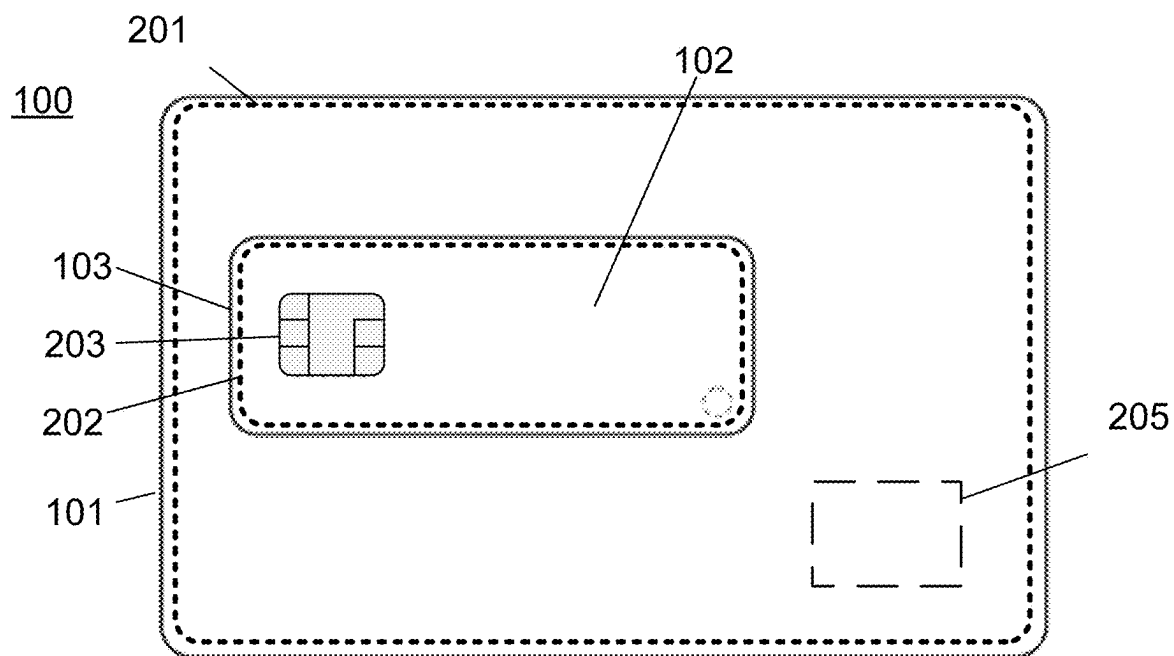


FIG. 2A

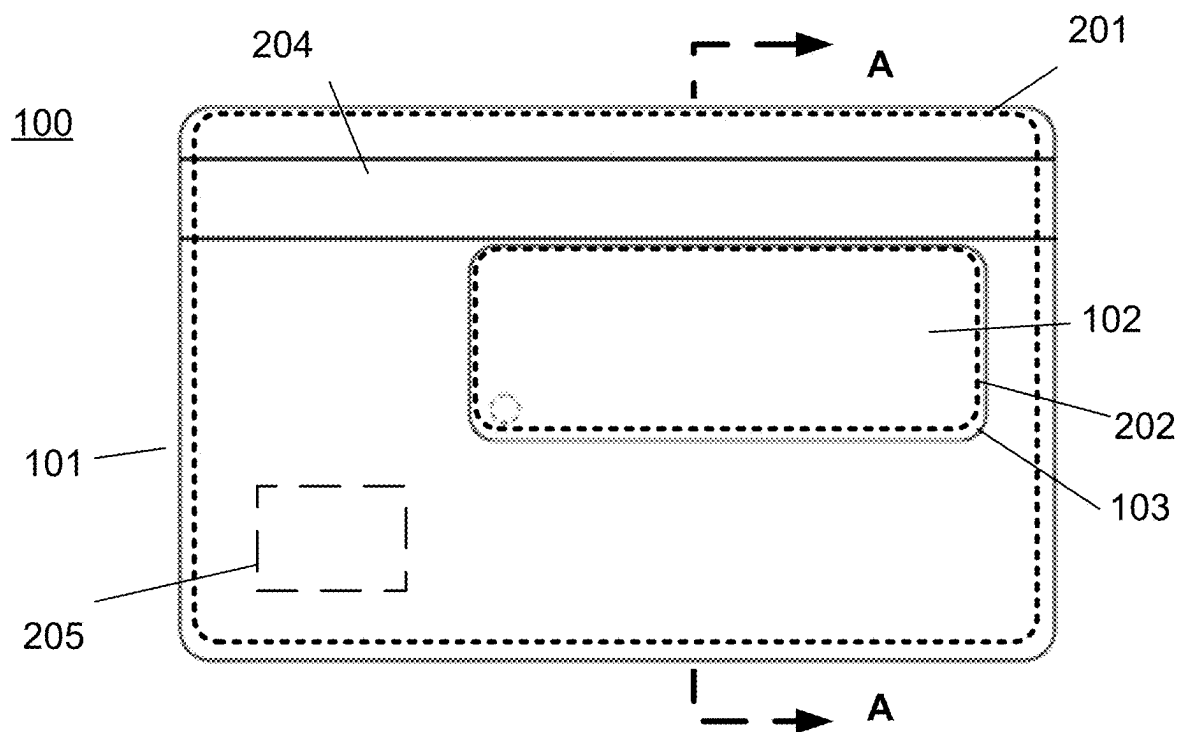


FIG. 2B

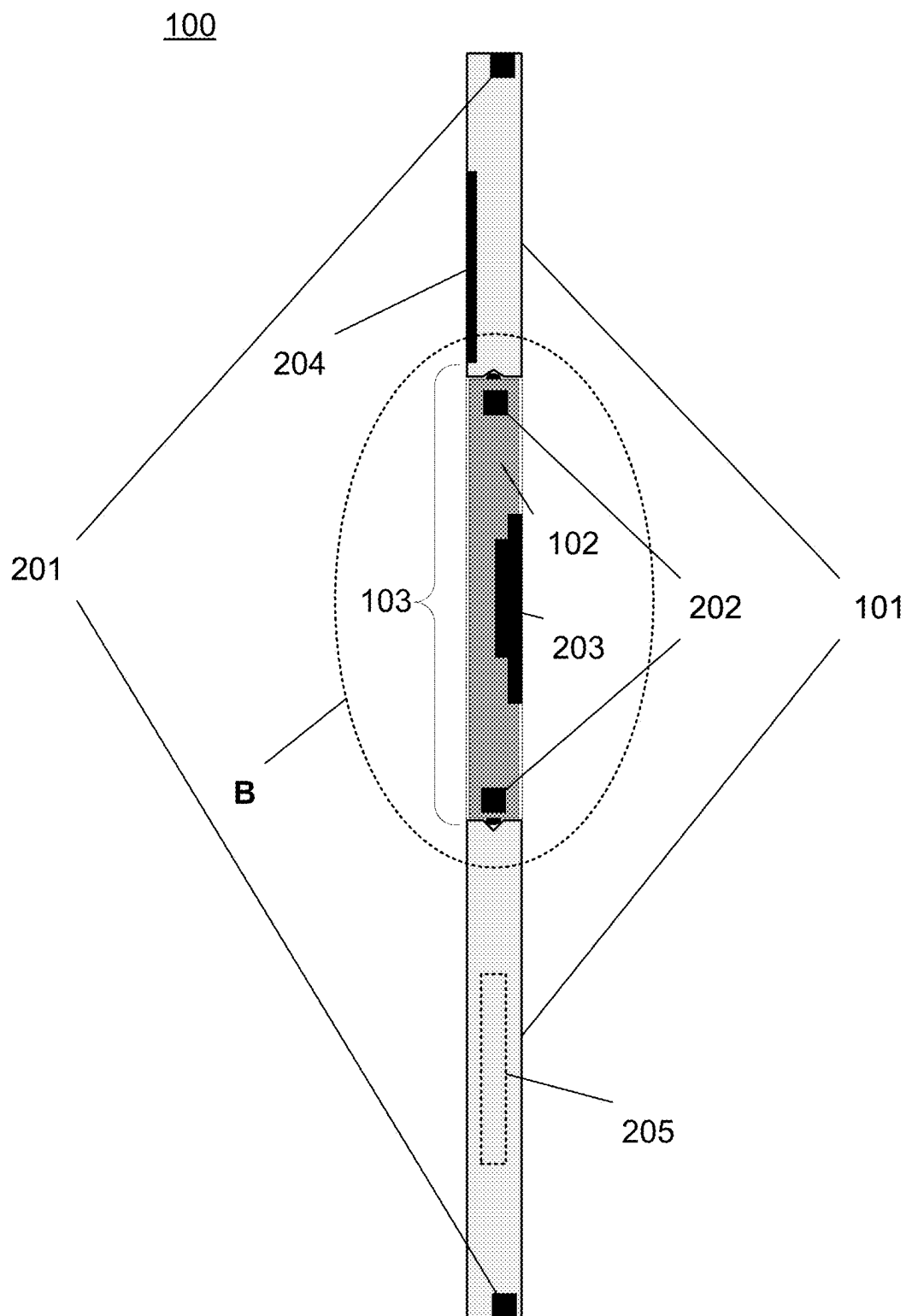
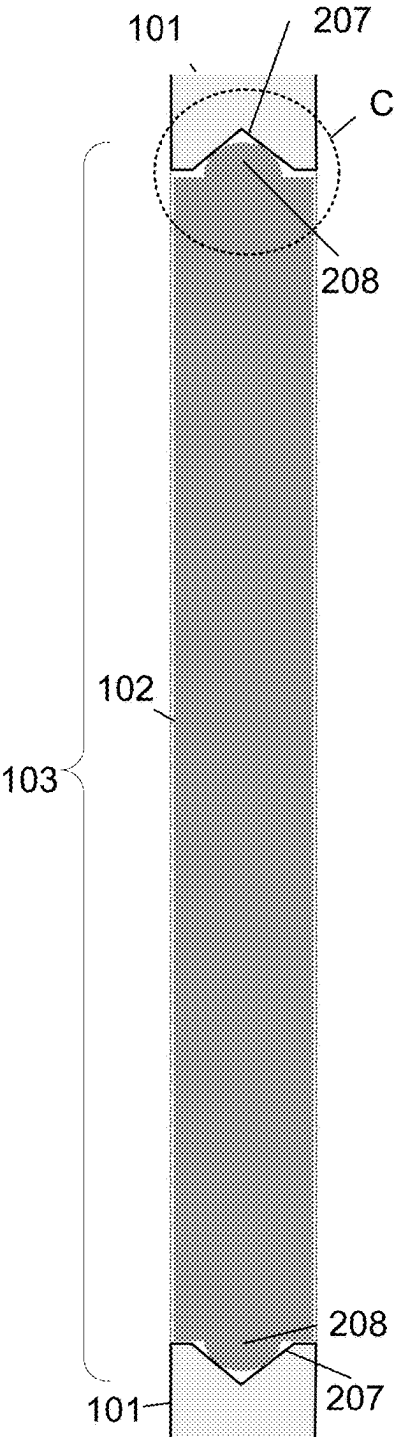


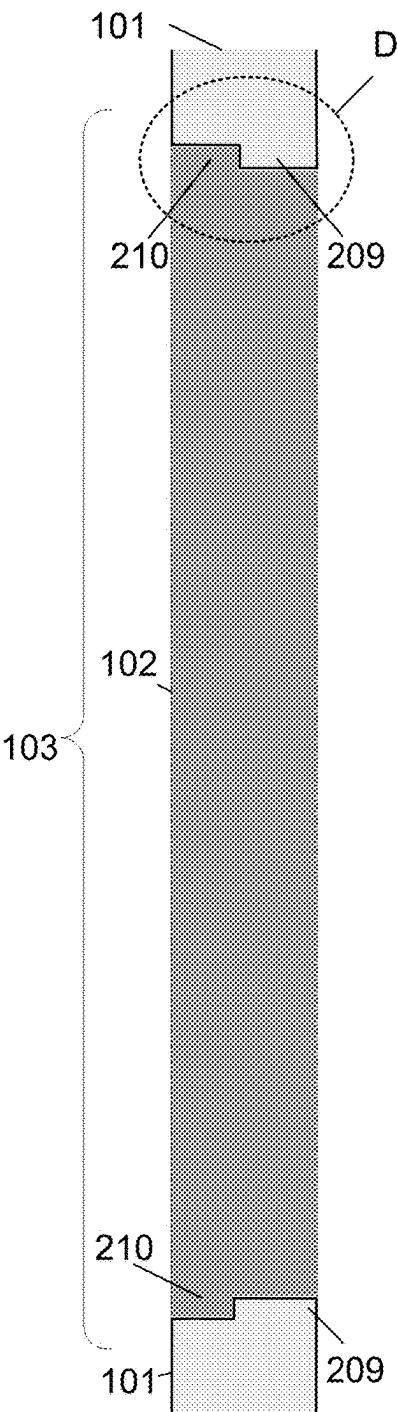
FIG. 2C

View A-A



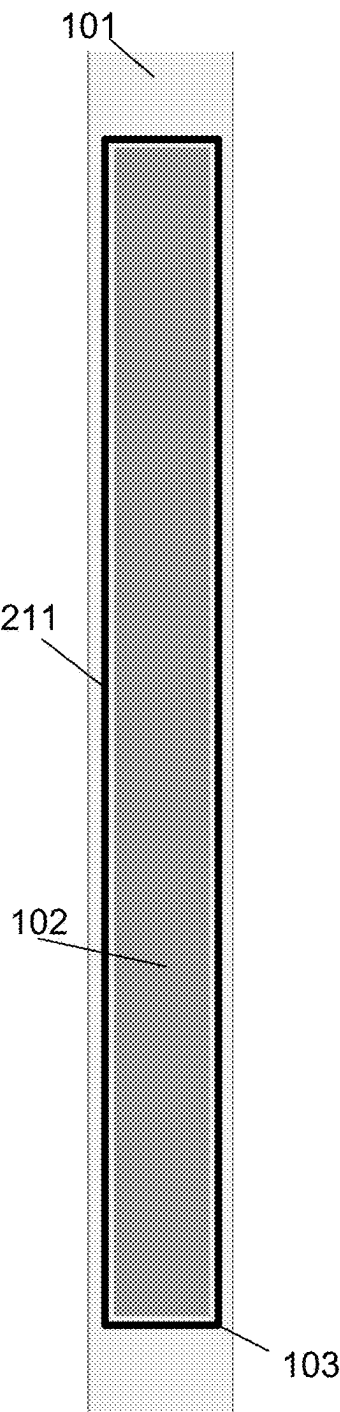
View B

FIG. 2D



View B

FIG. 2E



View B

FIG. 2F

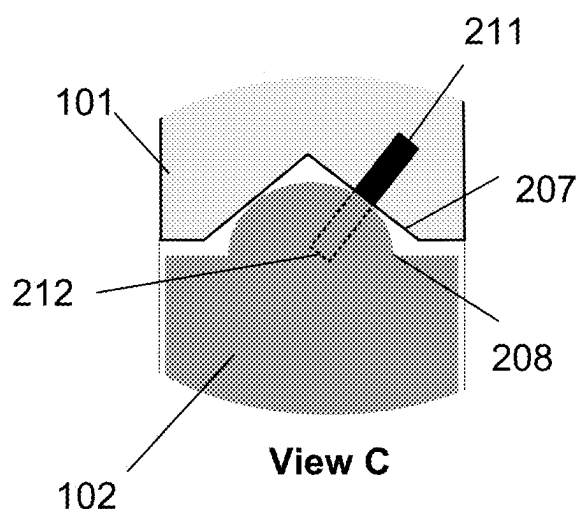


FIG. 2G

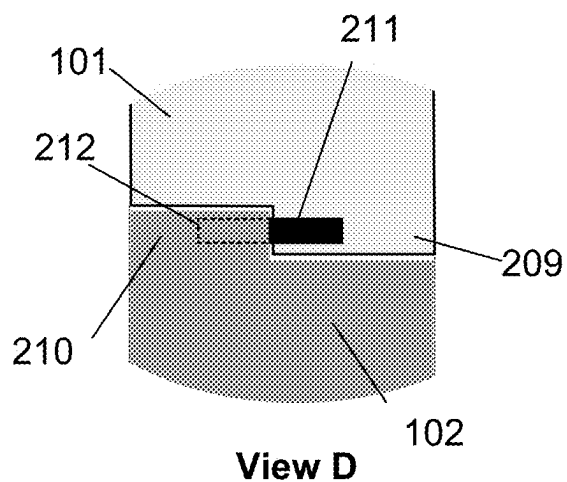


FIG. 2H

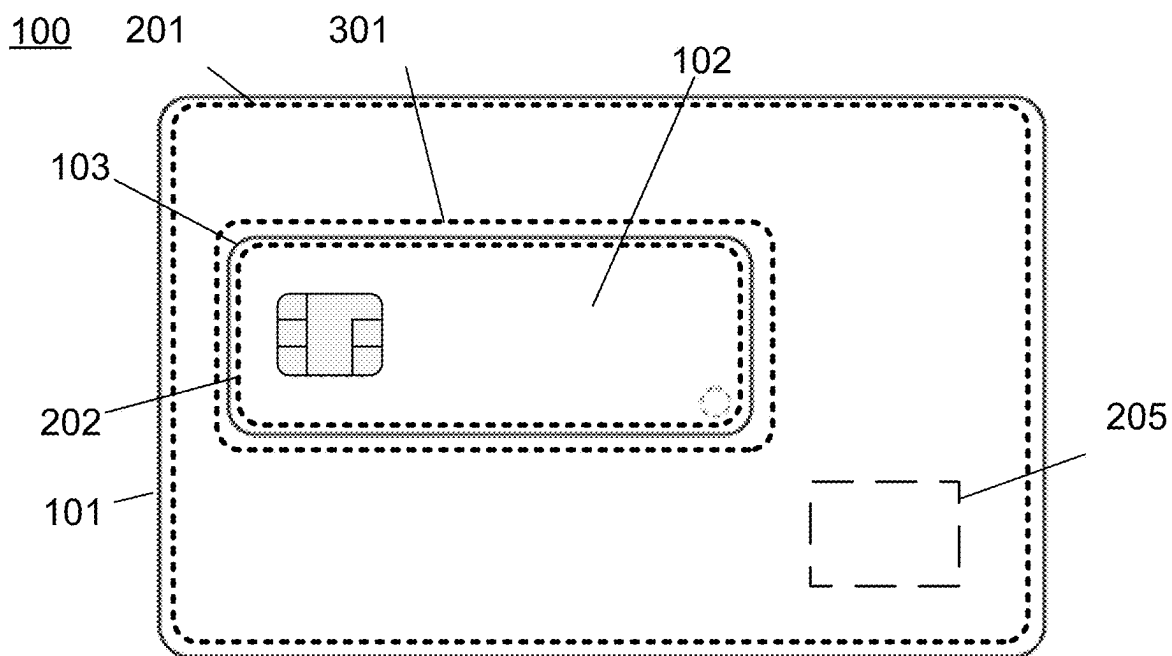


FIG. 3A

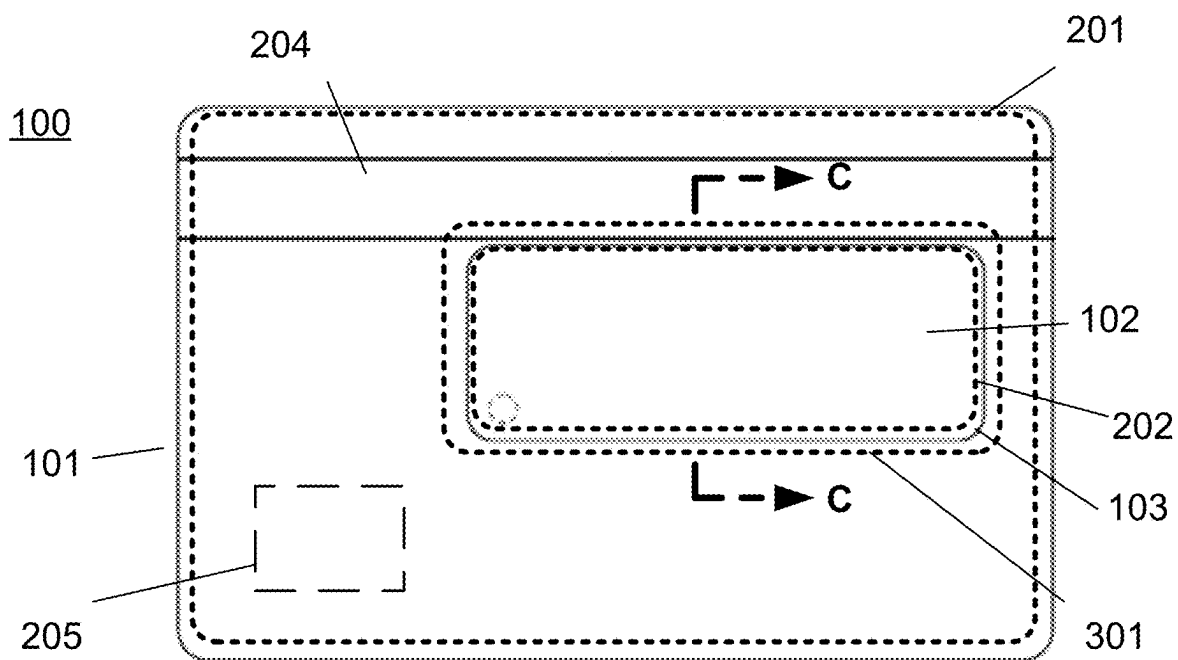


FIG. 3B

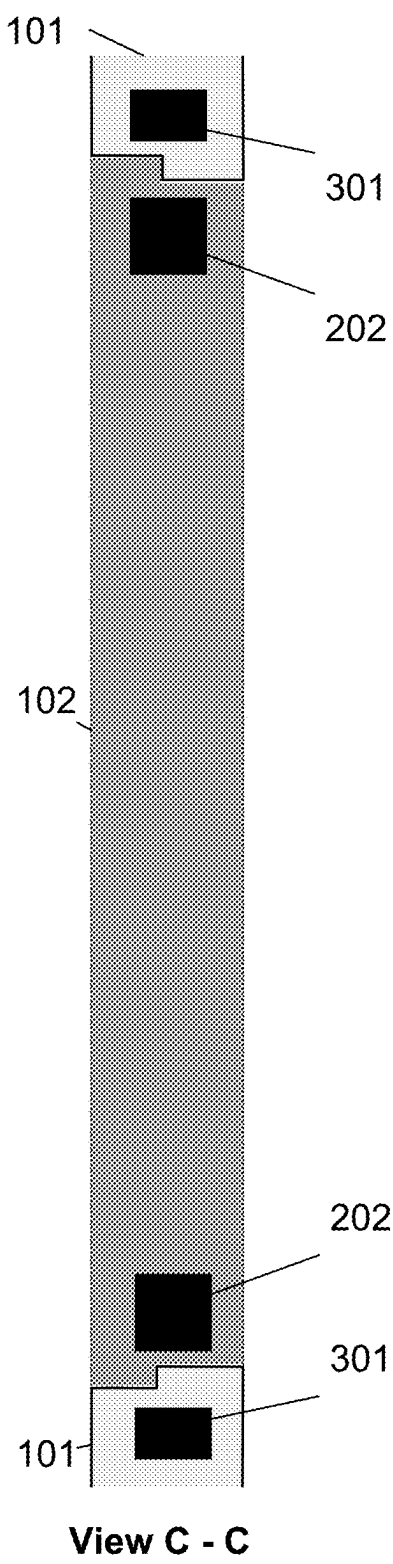


Fig. 3C

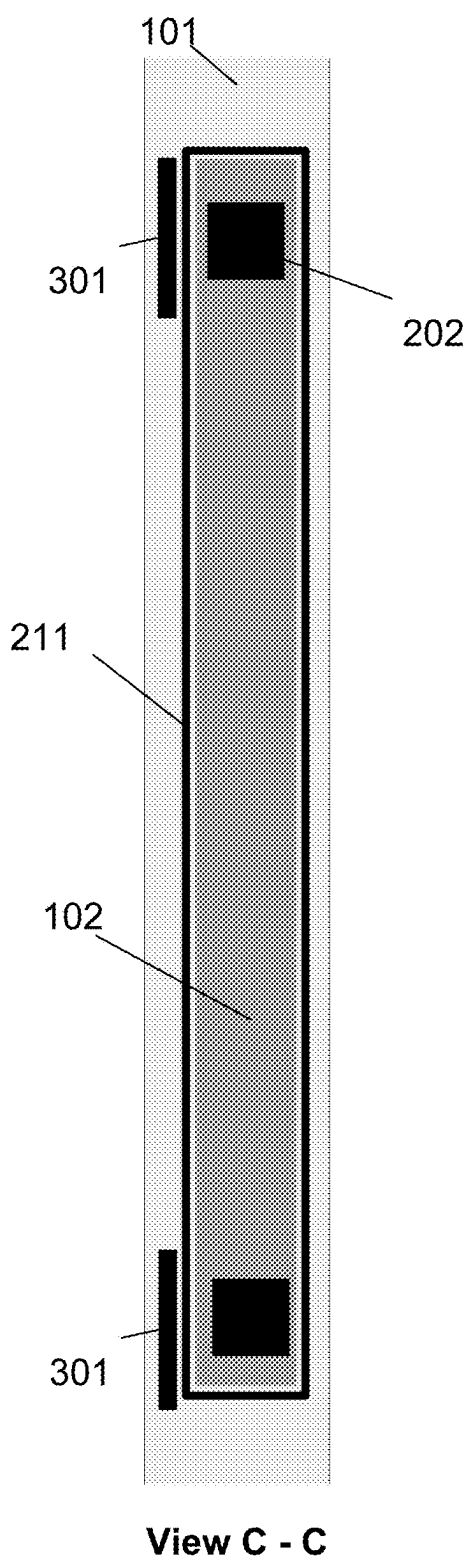
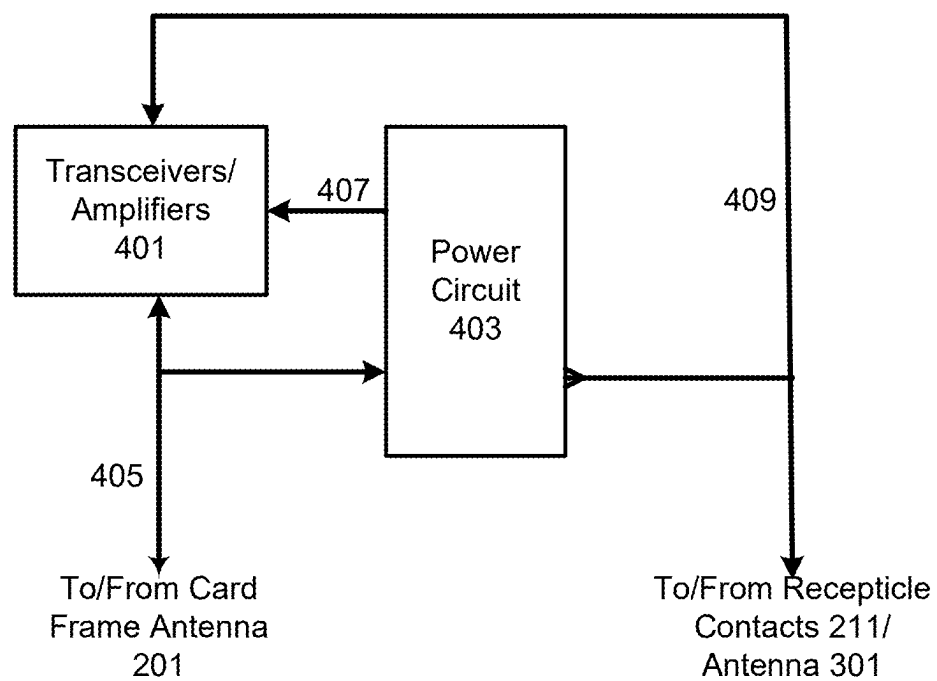


FIG. 3D

400**FIG. 4**

500

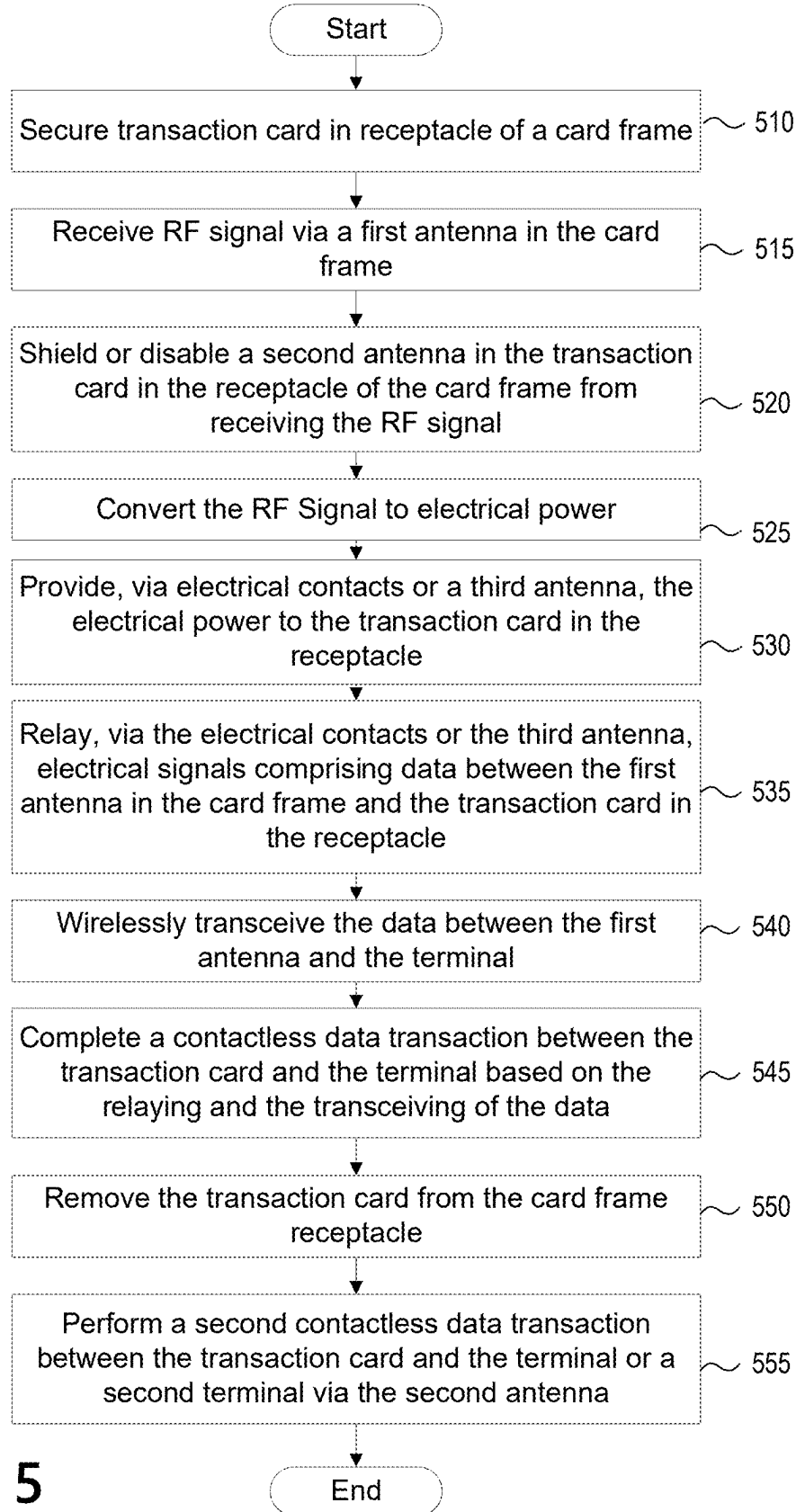
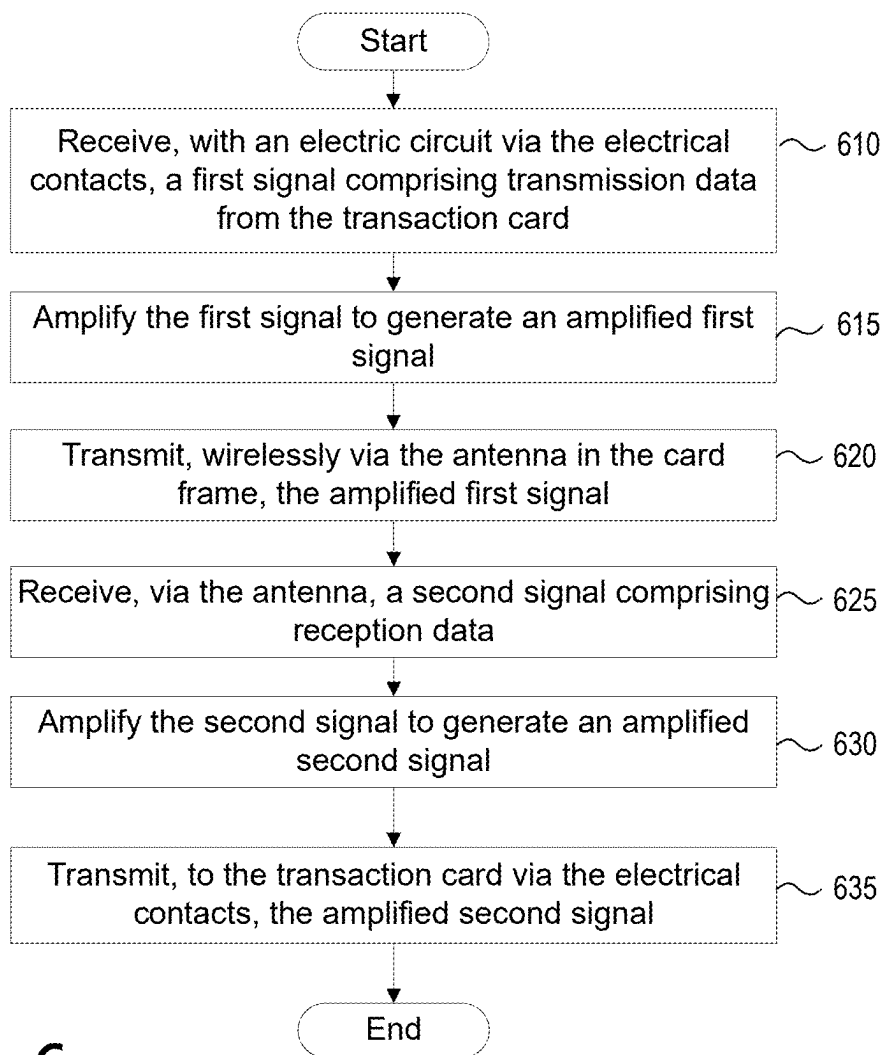


FIG. 5

600**FIG. 6**

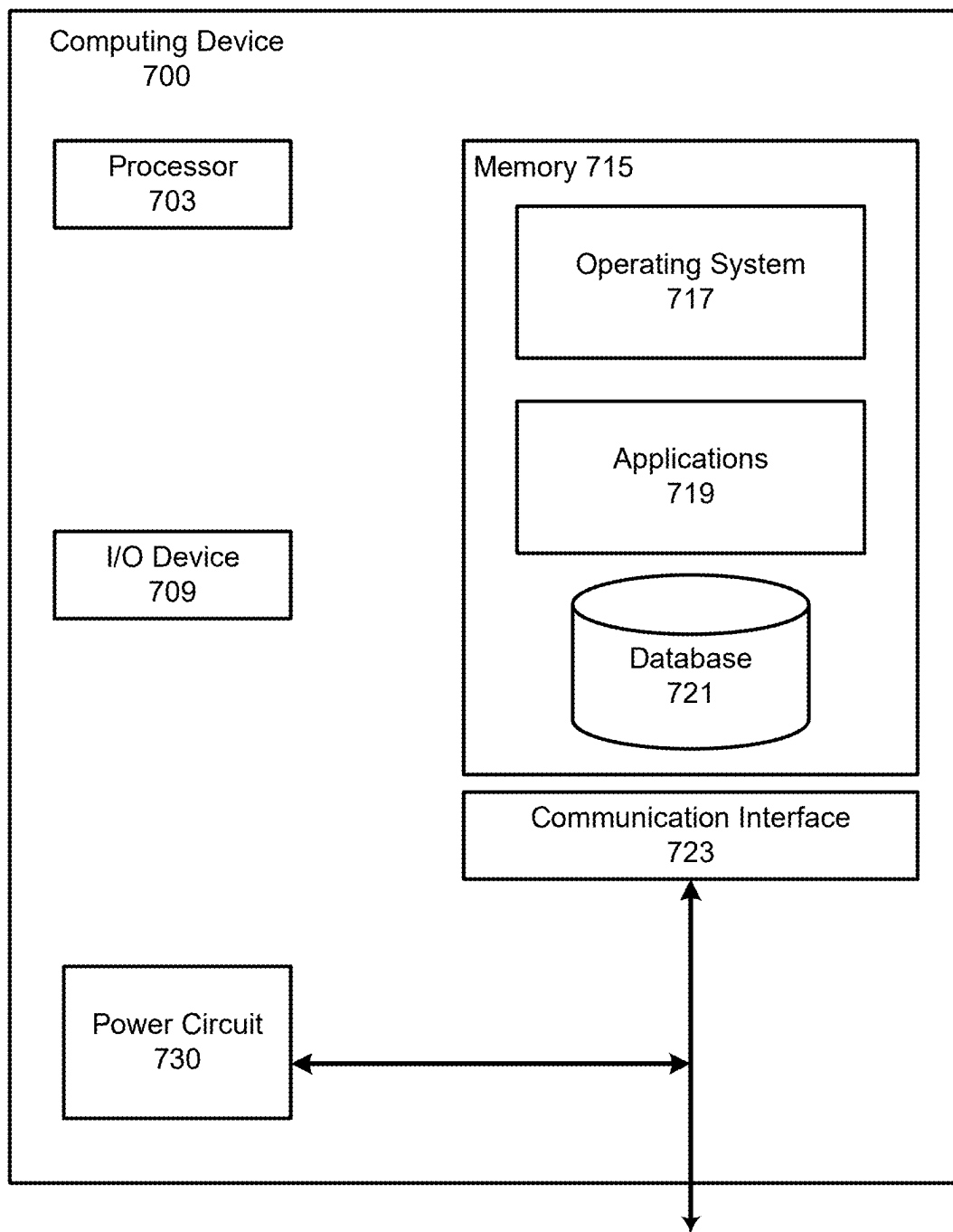


FIG. 7

To/From Antennas/Contacts

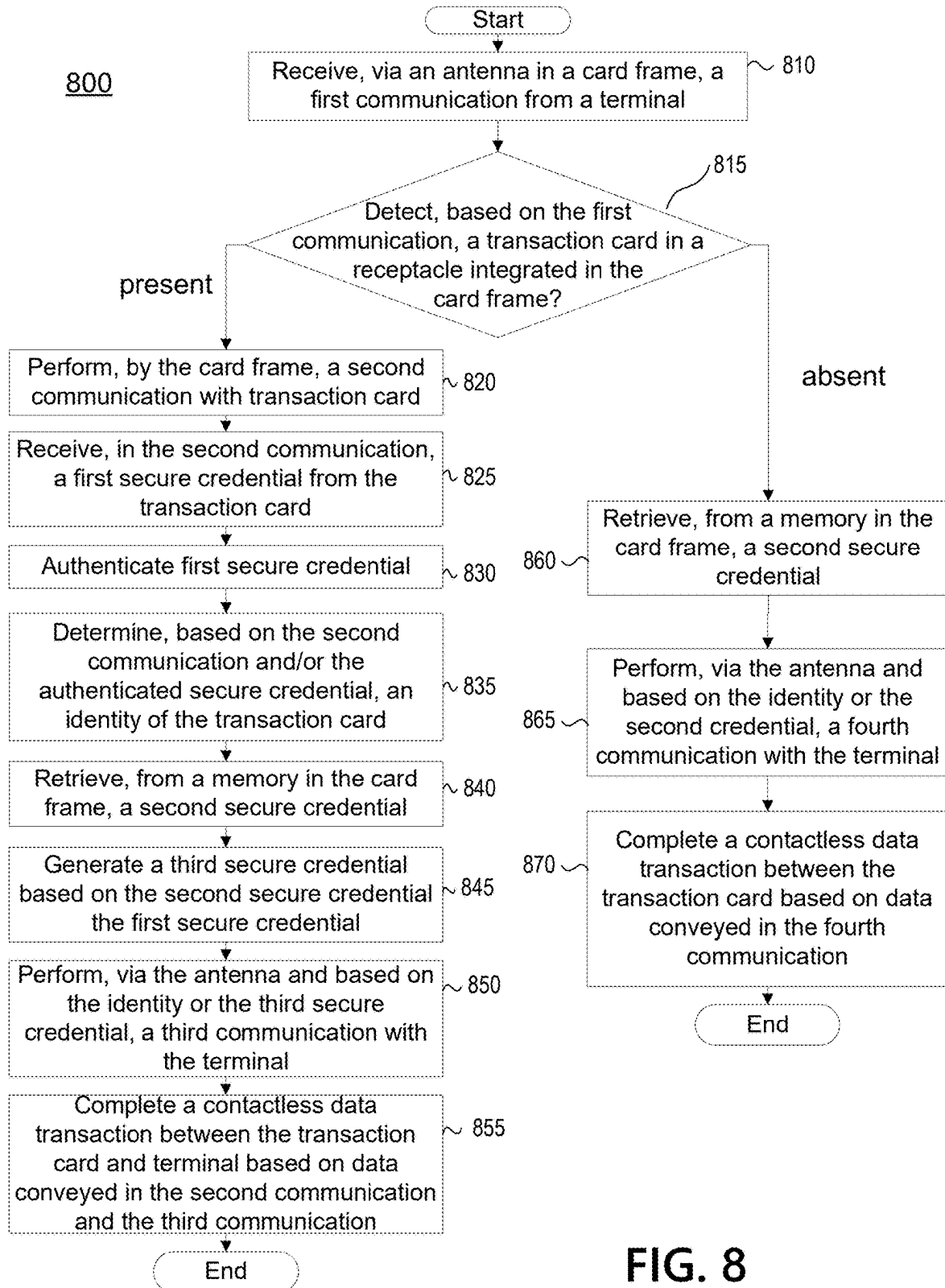


FIG. 8

TRANSACTION CARD ASSEMBLY

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of prior U.S. patent application Ser. No. 18/523,421, filed Nov. 29, 2023, which is a continuation-in-part application of U.S. patent application Ser. No. 17/539,633, filed Dec. 1, 2021, now U.S. Pat. No. 11,868,833, and U.S. patent application Ser. No. 17/539,636, filed Dec. 1, 2021, now U.S. Pat. No. 11,893,573, which are incorporated in their entireties by reference herein.

FIELD OF USE

[0002] Aspects of the disclosure relate generally to transaction cards and more specifically to a modular transaction card having multiple form factors.

BACKGROUND

[0003] Transaction cards have different form factors with different capabilities. Traditional credit cards, for example, may perform transactions contactlessly, using a magnetic strip, or via a smart chip. Transaction cards in smaller form factors, such as ones with a hole punch that can be attached to a keychain, are often more convenient to carry and may have the same transaction features, but not all card readers are able to work with the smaller dimensions.

SUMMARY

[0004] The following presents a simplified summary of various aspects described herein. This summary is not an extensive overview, and is not intended to identify key or critical elements or to delineate the scope of the claims. The following summary merely presents some concepts in a simplified form as an introductory prelude to the more detailed description provided below.

[0005] The methods, devices, systems, and/or computer-readable media disclosed herein relate to a transaction card assembly that includes a card frame having the traditional dimensions of a credit card, and a transaction card that is smaller than a traditional credit card (e.g., a “mini” transaction card). The transaction card may be capable of performing contactless data transactions (for example, credit card purchases) on its own via wireless communications, but may not be compatible with certain card readers, such as a chip reader, which accepts only a traditional credit card format. The card frame includes a receptacle that accepts and secures the transaction card and couples it to an antenna in the card frame, permitting the combined card frame and transaction card assembly to perform data transactions as a traditional form factor transaction card.

[0006] The receptacle in the card frame may have electrical contacts that mate with electrical contacts on the transaction card to couple it to the card frame antenna. In some variations, the card frame may have a second antenna that wirelessly communicates with an antenna in the transaction card when it is secured in the receptacle.

[0007] The card frame may include other features, such as electronics to provide power to the transaction card in the receptacle and may include shielding to prevent the antenna in the transaction card and the antenna in the card frame from both communicating with a card reader simultaneously.

[0008] In some variations, the card frame comprises a processing circuit for implementing a smart card frame. The processing circuit may include a computing device and memory storing computer instructions for enabling the card frame to work on its own as a transaction card without the smaller transaction card in the receptacle. The computing device may further implement authentication and cryptographic functions. For example, the card frame may authenticate the transaction card in the receptacle as a condition for completing a data transaction. The card frame may be used with multiple different transaction cards, with each combination of the card frame and different transaction cards having a unique identifier that is distinct from the identifier of each transaction card alone and different from the identifier of the card frame alone.

[0009] These features, along with many others, are discussed in greater detail below.

DESCRIPTION OF THE DRAWINGS

[0010] The present disclosure is described by way of example and not limited in the accompanying figures in which like reference numerals indicate similar elements and in which:

[0011] FIG. 1 illustrates a system in which a transaction card assembly may be used in accordance with one or more aspects of the disclosure;

[0012] FIGS. 2A-2H illustrate multiple views of a transaction card assembly in accordance with one or more aspects of the disclosure;

[0013] FIGS. 3A-3D illustrate multiple views of a transaction card assembly device in accordance with one or more aspects of the disclosure;

[0014] FIG. 4 illustrates a block diagram of an electrical circuit according to one or more aspects of the disclosure;

[0015] FIG. 5 illustrates a first example method for using a transaction card assembly to perform a data transaction according to one or more aspects of the disclosure.

[0016] FIG. 6 illustrates a second example method for using a transaction card assembly to perform a data transaction according to one or more aspects of the disclosure.

[0017] FIG. 7 illustrates a block diagram of a processing circuit according to one or more aspects of the disclosure; and

[0018] FIG. 8 illustrates a flow chart of a process for using a transaction card assembly to perform a data transaction according to one or more aspects of the disclosure.

DETAILED DESCRIPTION

[0019] In the following description of the various embodiments, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration various embodiments in which aspects of the disclosure may be practiced. It is to be understood that other embodiments may be utilized and structural and functional modifications may be made without departing from the scope of the present disclosure. Aspects of the disclosure are capable of other embodiments and of being practiced or being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. Rather, the phrases and terms used herein are to be given their broadest interpretation and meaning. The use of “including” and “comprising” and variations thereof is

meant to encompass the items listed thereafter and equivalents thereof as well as additional items and equivalents thereof.

[0020] By way of introduction, aspects discussed herein may relate to components, methods and techniques for a smart card system for performing contactless data transactions, the smart card system. The smart card system includes a smart card frame and one or more transaction cards. Each transaction card includes a first antenna and a first processor circuit that may perform, via the first antenna, a first contactless data transaction with a terminal. The first contactless data transaction may be based on a first secure credential stored in and unique to that transaction card. The smart card frame includes a flat sheet and a receptacle integrated in the flat sheet. The receptacle may be configured to secure, enable removal of, and resecure each of the one or more transaction cards in the smart card frame. The smart card frame may include a second antenna; and a second processor circuit integrated in the flat sheet. For each of the one or more transaction cards, the second processor circuit may be configured to generate a third secure credential based on a second secure credential stored in the smart card frame and based on the first secure credential of the transaction card. . . . The third secure credential may be unique to the combination of the transaction card and the smart card frame. The combined smart card frame and a transaction card may then perform a second contactless data transaction with the terminal based on the third secure credential.

[0021] FIG. 1 illustrates a system **10** that illustrates several components that may be found when conducting a transaction with different types of transaction cards. For example, system **10** shows a card reader **120** (e.g., a point-of-sale terminal), that may exchange data with transaction cards (e.g., **102**) through a plurality of communication techniques. The card reader **120** may be communicatively coupled to a server **140** via network **130**.

[0022] Card reader **120** may be any suitable card reader capable of exchanging data and/or information with transaction cards **102**. In this regard, card reader **120** may be a chip-based reader, a magnetic-based reader, an EM V reader, a wireless based reader, or any combination thereof. Accordingly, card reader **120** may include a display, a keypad, a network interface and a card interface. The display may present information to the cardholder, such as the amount owed, the status of the transaction, and whether the transaction has been approved or denied. A keypad or touch screen may allow a cardholder to input a personal identification number (PIN) code, password, an amount for withdrawal, and the like. A network interface may be a wired connection, wireless connection, a short-range wireless connection, a near field communication (NFC) connection, or any combination thereof. The network interface may permit card reader **120** to communicate with server **140**, via network **130**, for example, to authorize a transaction. The card interface may permit card reader **120** to communicate with transaction cards **102**. In these instances, card reader **120** may convey information related to the cardholder's account to transaction cards **102**. Card reader **120** may be limited in the ways it can communicate with different types of transaction cards. For example, card reader **120** may have a transaction card chip reader that only works with the dimensions of a standard size credit card, but not with the dimensions of transaction card **102**, which in some

instances, may have smaller or non-standard dimensions (e.g., a mini-card connectable to a key ring).

[0023] Various aspects described herein, which address this compatibility issue, are directed to a card assembly **100** comprising a card frame **101** that secures a smaller transaction card **102** in a receptacle **103**. In some embodiments, the card assembly **100** performs data transactions with card readers not compatible with transaction card **102**, and optionally, with a unique identifier that is distinct from an identifier of transaction card **102** when the transaction card is not secured in the card frame **101**.

[0024] Server **140** may be a stand-alone server, a corporate server, or a server located in a server farm or cloud-computer environment. According to some examples, server **140** may be a virtual server hosted on hardware capable of supporting a plurality of virtual servers. Server **140** may be configured to execute server-based software configured to provide cardholders with access to account information and perform routing banking functions. In some embodiments, the server-based software corresponds to client-based software executing on card reader **120**.

[0025] Network **130** may be any type of communications and/or computer network. The network **130** may include any type of communication mediums and/or may be based on any type of communication standards or protocols. In this regard, network **130** may include the Internet, a local area network (LAN), a wide area network (WAN), a wireless telecommunications network, and/or any other communication network or combination thereof.

[0026] Devices and systems **120**, **130**, and **140** in FIG. 1 may be implemented, in whole or in part, using one or more computing systems, for example, as described below with respect to FIG. 7.

[0027] Transaction card **102** and card frame **101**, individually or combined as card assembly **100**, may be configured to permit a cardholder to access one or more types of accounts. In this regard, transaction card **102**, card frame **101**, and/or card assembly **100** may behave as a credit card, a charge card, a debit card, a prepaid card, a smartcard, a payment card or an EM V card. In some embodiments, transaction card **102**, card frame **101**, and/or card assembly **100** may be an identification card, a club membership card, a rail pass card, or a building access card. As will be discussed in greater detail with respect to FIGS. 2 and 3, transaction card **102**, card frame **101**, and/or card assembly **100** may be chip-enabled and/or may include a magnetic strip. In further embodiments, transaction card **102**, card frame **101**, and/or card assembly **100** may include NFC capabilities, short-range wireless communication capabilities (e.g., Bluetooth®), wireless communication capabilities (e.g., Wi-Fi), or any combination thereof. The NFC capabilities, short-range wireless communication capabilities, and wireless communication capabilities may be referred to collectively as communication capabilities. These communication capabilities may permit transaction card **102**, card frame **101**, and/or card assembly **100** to communicate with card reader **120**.

[0028] Turning to FIG. 2A-2H, various views of card assembly **100** are illustrated. As illustrated in FIGS. 2A and 2B illustrating front and back views, card frame **101** of card assembly **100** may be of a standard size and made of a suitable substrate, such as plastic, metal, etc. For example, card frame **101** may be formed as a flat sheet having a rounded rectangle perimeter. In some variations, the flat

sheet is 3.361 to 3.382 inches wide, 2.119 to 2.133 inches high, and 0.027 to 0.033 inches thick.

[0029] Card frame **101** may include a magnetic strip **204** for storing data (e.g., credit card information) that may be read and written to by card reader **120**, and may include an antenna **201** capable of wireless communications (e.g., NFC, Bluetooth, Wi-Fi) with another device, such as carder reader **120** in FIG. 1.

[0030] Receptacle **103** in card frame **101** may be configured to secure, permit removal of, and resecure transaction card **102**. Card frame **101** may further include an electric circuit **205** for interfacing transaction card **102** to card frame antenna **201**, when transaction card **102** is secured in the receptacle **103**. Details of electric circuit **205** are further described below with respect to FIGS. 4 and 7.

[0031] Transaction card **102** may include a computer chip **203** and its own antenna **202** capable of wireless communications (e.g., NFC, Bluetooth, Wi-Fi) with another device, such as carder reader **120** in FIG. 1. As illustrated in the figures, antenna **202** and computer chip **203** may be embedded within transaction card **102**, and may be located anywhere in the perimeter of the transaction card **102** as illustrated in FIGS. 2A and 2B, and at any depth or on either surface of the transaction card **102**.

[0032] FIG. 2C illustrates a cross-sectional view A-A of FIG. 2B. As illustrated in FIG. 2C, the magnetic strip **204** may be embedded in, or on the surface of one side of, card frame **101**, enabling it to be read by a magnetic strip card reader. Antenna **201** and electric circuit **205** may also be embedded in, or on the surface of one side of card frame **101**. As illustrated in FIGS. 2A-2C, antenna **201** may be integrated along the perimeter of card frame **101**. In some variations, card frame **101** comprises a flat metal sheet and the antenna **201** is insulated from the metal sheet. In further variations, the antenna **201** is exposed along the edge of the card frame or is not completely surrounded by the metal sheet, so that the metal sheet does not interfere with reception and transmission of radio frequency communications by the antenna **201**.

[0033] The locations of magnetic strip **204**, antenna **201**, and electric circuit **205** are not limited to those locations illustrated in the figures and may be located anywhere in the perimeter of the card frame **101** illustrated in FIGS. 2A and 2B, and at any depth or on either surface of the card frame **101** illustrated in FIG. 2C.

[0034] As illustrated in FIGS. 2A-2C, in some variations transaction card **102**, when secured in receptacle **103**, may be substantially within the outer dimensions of card frame **101**, such that card frame **101** and transaction card **102** together form card assembly **100** as a uniform piece that appears and functions as a traditional transaction card (e.g., credit card).

[0035] Computer chip **203** in transaction card **102** may be a smart chip or integrated circuit. In this regard, chip **203** may include a microprocessor and memory, such as read only memory (ROM) and/or random access memory (RAM). Additionally, chip **203** may include one or more contact pads (illustrated in FIG. 2A) to receive electric power to operate the transaction card **102** and exchange signals with a terminal, such as card reader **120**. In some instances, the chip **203** may be configured to execute one or more applications. The applications may allow chip **203** to process payments. In other examples, the applications may allow the chip **203** to perform cryptographic processing,

authentication, define risk management parameters (e.g., when the transaction may be conducted offline), digitally sign payment data, and/or verify the cardholder. When secured in the receptacle **103** of the card frame **101**, the contact pads of chip **203** may be positioned to appear as contact pads for the card frame **101**. In some variations, transaction card **102** may be configured to perform a first transaction (e.g., a data transaction via chip **203** and/or antenna **202** authenticated with a first set of credentials) when the transaction card **102** is not inserted into card frame **101**, and perform a second transaction (e.g., a data transaction via chip **203** and/or antenna **201** authenticated with a second set of credentials) when the transaction card **102** is inserted into card frame **101**.

[0036] FIGS. 2D-2F illustrate various examples of view B of FIG. 2C to show in more detail the transaction card **102** inserted in receptacle **103**. As illustrated in FIGS. 2D and 2E, card frame **101** may be a flat sheet comprising two opposing surfaces separated by a thickness and bounded by a perimeter (illustrated in FIGS. 1A and 1B), wherein the receptacle **103** comprises a hole passing completely through the thickness of the flat sheet. The receptacle **103** may have a perimeter (as illustrated in FIGS. 2A and 2B) that matches entirely, or only at some edges of, the perimeter of the transaction card **102**. As illustrated in FIG. 2D, the transaction card **102** and receptacle **103** may have an interference fit in which the profile of the transaction card **102** has a protrusion **208**, which fits within a groove **207** of the receptacle **103** to secure the transaction card **102** in the card frame **101**. The material of the protrusion **208** and/or the walls of the groove **207** may be flexible to allow the transaction card **102** to be secured, removed, and resecured in the receptacle **103** with the application of opposing forces perpendicular to the faces of the card frame **101** and transaction card **102** respectively, for example, to snap the transaction card **102** into the receptacle **103**. While protrusion **208** is illustrated as curved and groove **207** is illustrated in the shape of a “v,” these may be of any profile that provides an interference fit. Additionally, the profiles may be reversed so that the perimeter of the transaction card **102** has a groove, and the receptacle **103** has a protrusion.

[0037] FIG. 2E illustrates another example, in which the perimeters of the receptacle **103** and transaction card **102** have mating rims **210** and **209** respectively with mirrored profiles. In some variations, the transaction card **102** may be inserted from only one side of the card frame **101**. The transaction card **102** may be secured in the receptacle **103** by friction between the perimeters of the transaction card **102** and receptacle **103**. In other variations, the transaction card **102** may be magnetically coupled to the receptacle **103** and/or card frame **101** to secure, enable removal of, and resecure the transaction card **102** in the card frame **101**. Each of these configurations can be used together, for example by some edges of the transaction card **102** being secured with a groove/protrusion, and some with mirrored rims that are coupled magnetically. Transaction card **102** may be configured to detect whether it is inserted in the card frame **101**, and based on this detection, perform different operations (e.g., perform different types of data transactions or take on different identities).

[0038] While FIGS. 2D and 2E illustrate the receptacle **103** as a hole passing completely through the flat sheet of the card frame, in other variations, the receptacle **103** may be a

recess in one of the two opposing surfaces with the other surface being completely or partially closed.

[0039] FIG. 2F illustrates another variation of receptacle 103 comprising a slot 211 having an open end, along the perimeter between the two opposing surfaces of the card frame 101, through which the receptacle 103 is configured to secure, enable the removal of, and resecure the transaction card 102.

[0040] In some variations, the card frame 101 and/or receptacle 103 may provide an electromagnetic shield preventing the antenna 202 in the transaction card 102 from receiving or transmitting radio frequency signals while the transaction card 102 is secured in the receptacle 103. For example, in the receptacle 103 in FIG. 2F, one or both surfaces of the card frame 101 may be coated or made of a conductive material (e.g., aluminum, stainless steel, titanium), which covers the transaction card 102 partially or completely. In this way, the card frame 101 may disable the transaction card antenna 202 while the transaction card 102 is secured in the receptacle 103, thus preventing both antennas 201 and 202 from relaying data transactions simultaneously. Alternatively or additionally, transaction card 102 may disable its antenna 202 based on detecting that the transaction card 102 is secured in the receptacle 103.

[0041] As previously discussed, when transaction card 102 is secured in receptacle 103, it may be interfaced to antenna 201 in the card frame 101 via electric circuit 205. FIGS. 2G-2H illustrate views C and D of FIGS. 2D-2E, respectively, which illustrate details of electrical contacts for electrically coupling transaction card 102 to electric circuit 205. As illustrated in these figures, card frame 101 may include one or more electrical contacts 211 along the perimeter of the receptacle 103 that contact a corresponding one or more electrical contacts 212 on the perimeter of the transaction card 102 when the transaction card 102 is secured in the receptacle 103. In FIG. 2G, contacts 211 and 212 are illustrated on the mating protrusion 208 and groove 207, respectively, but the contact coupling can be positioned at any location at which the transaction card 102 comes into contact with the card frame 101 so that contacts 211 and 212 touch. Similarly, in FIG. 2H, contacts 211 and 212 are illustrated on the first rim of card frame 101 and mating second rim of transaction card 102, respectively, but the contact coupling can be positioned at any location at which the transaction card 102 comes into contact with the card frame 101 so that contacts 211 and 212 touch. For example, if the receptacle 103 is a recess or a slot, the card frame 101 may have contacts on the bottom surface of the recess or inside surface of the slot, which contact corresponding contacts on a surface of the transaction card 102.

[0042] FIGS. 3A-3D illustrate different views of another variation of card assembly 100, in which card frame 101 communicates with transaction card 102 wirelessly (e.g., without using contacts 211 and 212). FIG. 3A illustrates a front view, FIG. 3B illustrates a back view, and FIGS. 3C and 3D illustrate cross-section views C-C of receptacle 300. As illustrated in these figures, card frame 101 includes an additional antenna 301 proximate to the receptacle 103. For example, as illustrated in FIGS. 3A, 3B, and 3C, antenna 301 may be embedded in the card frame 101 and may encircle the perimeter of the receptacle 103, and thus encircle antenna 202 when the transaction card 102 is secured in the receptacle 103. In other variations, for example when the receptacle 103 comprises a recess or a slot

as illustrated in FIG. 3D, antenna 301 may be embedded in or on the surface of the wall of the slot or on the bottom of a recess. In this assembly, transaction card 102 and card frame 101 exchange data via radio frequency communication between antennas 202 and 301, which may include implementing a wireless protocol (e.g., NFC, Wi-Fi, Bluetooth®, and/or Bluetooth Low Energy (BLE)). In some variations, antennas 301 and 202 provide inductive power transfer between card frame 101 and transaction card 102. In some variations the card frame 101 includes both electrical contacts 211 and antenna 301, which may alternatively be used, or used in combination, depending upon whether the transaction card 102 has corresponding features and capabilities (e.g., contacts 212 and antenna 202).

[0043] Transaction card 102 may be coupled to antenna 201 in the card frame by electric circuit 205 via the electrical contacts 211 and 212 or by the electromagnetically coupled antennas 202 and 301. In one variation, electric circuit 205 may comprise wire conductors and (optionally) passive components (e.g., capacitors, resistors, inductors) that electrically (e.g., directly or capacitively) connect antenna 201 to contacts 211 and/or antenna 301.

[0044] FIG. 4 illustrates circuit 400, which is another variation of electric circuit 205. Circuit 400 may include one or more conductors 405 that are connected between antenna 201 and transceivers and/or amplifiers 401 in card frame 101. Antenna 201 may receive and radiate radio frequency signals, which correspond to signals carried on the one or more conductors 405 to and from transceivers and/or amplifiers 401. Similarly, one or more conductors 409 are connected and carry electrical signals between contacts 211 or antenna 301 and transceivers and/or amplifiers 401. The transceivers and/or amplifiers 401 condition the signals, for example by amplifying and filtering them, and exchange the conditioned signals between conductors 405 and 409 to provide a complete communication path for data carried in the signals between the antenna 201 in the card frame and the transaction card 102 in the receptacle 103.

[0045] For example, the electric circuit 400 may be configured to receive, via contacts 211 and conductors 409, a first signal comprising transmission data from the transaction card 102 (through contacts 212), amplify the first signal with transceivers and/or amplifiers 401 to generate an amplified first signal, and transmit wirelessly the amplified first signal including the transmission data via conductors 405 and antenna 201. Similarly, the electric circuit 400 may be configured to receive wirelessly, via antenna 201 and conductors 405, a second signal comprising reception data, amplify the second signal to generate an amplified second signal with transceivers and/or amplifiers 401; and transmit the amplified second signal including the reception data, to the transaction card 102 via conductors 409 and contacts 211. This relaying of data between the antenna 201 in the card frame 101 and the transaction card 102 in the receptacle 103, and wirelessly transceiving the data between the antenna 201 and a terminal may be used to perform a contactless data transaction between the transaction card 102 and a terminal. In some variations, transceivers and/or amplifiers 401 may include communication protocol capabilities, such as NFC, Wi-Fi, Bluetooth®, and/or BLE.

[0046] Electric circuit 400 may further include a power circuit 403, which is configured to generate electric power from signals from antenna 201 in the card frame 101. For example, radio frequency electromagnetic energy (e.g.,

radio frequency wireless signals) may be received by antenna 201 and conducted along conductors 405 to power circuit 403. These may be the same or different signals that carry data and are conditioned and amplified by transceivers/amplifiers 401. Power circuit 403 may include a power converter (for example, comprising a capacitor and a diode) that converts the radio frequency signals to electrical power (e.g., alternating current or direct current power). The generated electrical power may be provided via conductors 407 to energize the electrical circuits within the transceivers and/or amplifiers 401. The electrical power may additionally or alternatively be provided via conductors 409 to contacts 211 and 212 to the transaction card 102. In some variations, the electrical power is converted back to radio-frequency signals and transmitted via antenna 301 to the transaction card 102, which may be configured to receive these signals via antenna 202 and convert them to electrical power internally in the transaction card 102 (e.g., inductive charging).

[0047] FIG. 5 illustrates an example method 500 for using the transaction card 102 with and optionally without card frame 101. In step 510, transaction card 102 may be secured in receptacle 103 of card frame 101 as described above (e.g., with an interference fit or magnetic coupling). In step 515, radio frequency electromagnetic energy (e.g., a wireless radio frequency signal) may be received via antenna 201.

[0048] In step 520, second antenna 202 in the transaction card 102 is disabled from receiving radio frequency signals from outside of the card frame 101. In some examples, this prevents the card assembly 100 (101 and 102 together) from performing or attempting to perform multiple transactions with a card reader (e.g., a point-of-sale terminal), by for example, receiving radio frequency transmission on both antenna 201 and 202. In some variations (for example, as shown in FIG. 2F and described above), antenna 202 is disabled by the card frame 101 by providing an electromagnetic shield around the antenna 202. In other variations, the transaction card 102 detects that it is secured in the receptacle 102, and based on this detection, disables the antenna 202 internally in the transaction card 102.

[0049] In step 525, the card frame 101 converts the received radio frequency signals into electrical power (e.g., direct-current or alternating-current power), and in step 530, the electrical power is provided via electrical contacts 211/212 or antennas 301 and 202 (via inductive coupling) to the transaction card 102 in the receptacle 103 as discussed above.

[0050] In step 535, the card frame 101 may relay, via the electrical contacts 211 and 212 or antennas 301 and 202, electrical signals comprising data between antenna 201 in the card frame 101 and the transaction card 102 in the receptacle 103. This may be performed by electric circuit 400 as previously discussed, or by computing device 700, which is further described below with respect to FIG. 7. In step 540, card frame 101 may wirelessly transceive this data to and from a card reader 120 (e.g., a terminal). The transceiving may include radiating and/or receiving the data in radio frequency signals from antenna 201. In step 545, the card assembly 100 completes a contactless data transaction between the transaction card 102 and the terminal based on the relaying and the transceiving of the data.

[0051] In step 550, the transaction card 102 is removed from the card frame receptacle 103, as previously described above. Once removed, in step 550, the transaction card 102

may perform a second contactless data transaction with the card reader (or a different card reader) (e.g., terminal) using its antenna 202. The steps of process 500 may be performed in other orders and all steps need not be performed.

[0052] FIG. 6 illustrates a method 600 for card frame 101, for example using electric circuit 400, to relay and transceive data. In step 610, the card frame 101 may receive (for example via the electrical contacts 211 or antenna 301) a first signal comprising transmission data from the transaction card 102. In step 615, the card frame 101 may amplify (for example using transceivers and/or amplifiers 401) the first signal to generate an amplified first signal. In step 620, the card frame 101 may transmit, wirelessly via antenna 201, the amplified first signal, for example to a card reader 120. In step 625, the card frame 101 may receive, via antenna 201, a second signal comprising reception data, and in step 630, the card frame 101 may amplify (for example using transceivers and/or amplifiers 401) the second signal to generate an amplified second signal. In step 635 the card frame 101 may transmit (for example via the electrical contacts 211 or antenna 301) the amplified second signal to the transaction card 102. The transmission of the first signal comprising transmission data and the reception of the second signal comprising reception data may be performed in any order and may be related, with one being based on, or in response to, the other, and with both part of a contactless data transaction (e.g., a credit card transaction). Processes 500 and 600 may be performed separately or together.

[0053] In some variations of card frame 101, electric circuit 205 includes a processing circuit for implementing a smart card frame. For example, electric circuit 205 may comprise computing device 700 as illustrated in FIG. 7. Computing device 700 may include a processor 703 for controlling overall operation of the computing device 700 and its associated components, input/output device 709, memory 715, and/or communication interface 723. A data bus may interconnect processor(s) 703, memory 715, I/O device 709, and/or communication interface 723.

[0054] Input/output (I/O) device 709 may include a port (e.g., contacts, conductors, modem) through which the computing device 700 may receive input, such as for initial programming, receiving authentication keys, etc., prior to being issued to a cardholder.

[0055] Software may be stored within memory 715 to provide instructions to processor 703 allowing computing device 700 to perform various actions. For example, memory 715 may store software used by the computing device 700, such as an operating system 717, application programs 719, and/or an associated internal database 721. The various hardware memory units in memory 715 may include volatile and nonvolatile media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data. Memory 715 may include one or more physical persistent memory devices and/or one or more non-persistent memory devices. Memory 715 may include, but is not limited to, RAM, ROM, electronically erasable programmable read only memory (EEPROM), flash memory or other memory technology that may store information and that may be accessed by processor 703.

[0056] Communication interface 723 may include one or more transceivers, amplifiers, digital signal processors, and/or additional circuitry and software for communicating via antennas 201 and/or 301 and/or contacts 211. Communica-

tion interface **723** may also include near field communication (NFC) capabilities, short-range wireless communication capabilities (e.g., Bluetooth®), wireless communication capabilities (e.g., Wi-Fi), or any combination thereof. Communication interface **723** may include some or all of the features of electric circuit **400** illustrated in FIG. **4**.

[0057] Computing device **700** may further include a power circuit **730**, which may be the same as power circuit **403** described with respect to FIG. **4** for converting radio frequency electromagnetic signals to electrical power for powering computing device **700** and transaction card **102** as previously described.

[0058] Processor **703** may include a single central processing unit (CPU), which may be a single-core or multi-core processor, or may include multiple CPUs. Processor(s) **703** and associated components may allow the computing device **700** to execute a series of computer-readable instructions to perform some or all of the processes described herein. Although not illustrated in FIG. **7**, various elements within memory **715** or other components in computing device **700**, may include one or more caches, for example, CPU caches used by the processor **703**, page caches used by the operating system **717**, and/or database caches used to cache content from database **721**. For embodiments including a CPU cache, the CPU cache may be used by one or more processors **703** to reduce memory latency and access time. A processor **703** may retrieve data from or write data to the CPU cache rather than reading/writing to memory **715**, which may improve the speed of these operations.

[0059] Although various components of computing device **700** are described separately, functionality of the various components may be combined and/or performed by a single component and/or multiple computing devices in communication. And although various components of computing device **700** are described separately from the various components of electric circuit **400**, these various components and their functionality may be combined and/or performed by a single component and/or multiple computing devices in communication.

[0060] The inclusion of a processing circuit, such as computing device **700**, greatly expands the capabilities of card frame **101**, such as: enabling it to work as a transaction card on its own (without transaction card **102** secured in the receptacle), providing security measures limiting the use of the card frame to only certain paired transaction cards **102**, and enabling the card frame to take on multiple different identities, depending upon whether a transaction card is inserted in the receptacle and depending upon which of multiple different transaction cards is inserted into the receptacle.

[0061] FIG. **8** illustrates a method **800** for using the card frame **101** that includes a processing circuit, such as computing device **700**. Process **800** begins with step **810**, in which the card frame **101** receives, via antenna **201**, a first communication from a terminal, such as card reader **120**. The first communication may communicate data (in either or both directions) and be the beginning or part of a data transaction (e.g., a contactless transaction, NFC transaction) with the terminal. In step **815** the card frame **101** detects, based on the first communication, whether a transaction card **102** is secured in receptacle **103**. The detection may, for example, be based on a communication between the transaction card **102** and computer device **700**, or may be based on a measurement of an electrical parameter (e.g., detection

of a resistance at contacts **211**). In response to detecting that the transaction card **102** is present in the receptacle **103**, steps **820-855** may be performed to complete a contactless data transaction based on the combination of card frame **101** and transaction card **102** together as card assembly **100**. In response to detecting that the transaction card **102** is absent from the receptacle **103**, steps **860-875** may be performed to complete a contactless data transaction based on the card frame **101** alone.

[0062] If transaction card **102** is present in the receptacle **103**, card frame **101** in step **820**, performs a second communication with transaction card **102**. The second communication may convey data (in either or both directions) for performing the data transaction with the terminal. The card frame **101** may in step **825** receive in the second communication, a first secure credential from transaction card **102**, which may be unique to transaction card **102**, and may in step **830** authenticate the first secure credential, for example, using a decryption and/or authentication application executed in computing device **700**. In step **835**, the card frame **101** (for example, using computing device **700**) may determine, based on the second communication and/or the authenticated secure credential, an identity of transaction card **102**.

[0063] In step **840** the computing device **700** may (optionally) retrieve from a memory (e.g., **715**) in the card frame **101**, a second secure credential uniquely associated with the card frame **101**. Performance of step **840** may be based on confirming that the identity or authenticated secure credential of the transaction card **102** is authorized to be used with the card frame **101**. For example, computing device **700** may have stored in memory, a list of one or more identities of different transaction cards authorized to be used with the card frame **101**. If the transaction card **102** is not authorized to be used with the card frame **101** (e.g., because transaction card **102** is not in the list), the process may end without completing the data transaction.

[0064] In step **845** computing device **700** in the card frame **101** may generate a third secure credential based on the first secure credential and (optionally) based on the second secure credential. The third secure credential may be unique to the combination of the card frame **101** and transaction card **102** (for example, by being derived from the first and second secure credentials). Computing device **700** may be configured to generate multiple different third secure credentials based on the second secure credential and, respectively, multiple different first secure credentials of multiple different transaction cards **102**.

[0065] In step **850** card frame **101** may perform, via antenna **201** and based on the identity of the transaction card **102** in the receptacle or based the third secure credential, a third communication with the terminal. The performance of steps **845** and/or **850** may be based on or in response to the successful authentication of the first secure credential. The third communication may contain data conveyed in the second communication and additional data (e.g., the third secure credential). In step **855** the card frame **101** may complete a contactless data transaction between the transaction card **102** in the receptacle and the terminal based on data conveyed in the second communication and the third communication. In each of the communications, the data may be conveyed (in either or both directions) and (optionally) encrypted, with computing device **700** performing encryption and decryption of the data.

[0066] Returning to step 815, if the transaction card 102 was determined to be absent from the receptacle 103 integrated in the card frame 101, step 860 may be performed in which the card frame 101 retrieves from the memory in the card frame 101, the second secure credential as described above with respect to step 840. In step 865 card frame 101 may perform, via antenna 201 and based on the second secure credential, a fourth communication with the terminal. In this step, the second secure credential is uniquely associated with just the card frame 101 and is distinct from the secure credentials of the transaction cards 102. In step 875 card frame 101 completes a data transaction (e.g., contactless data transaction, NFC transaction) between the card frame 101 and the terminal based on data conveyed (in either or both directions) in the fourth communication. In the fourth communication, the data may be encrypted, with computing device 700 performing encryption and decryption of the data.

[0067] With the steps of process 800, the card frame 101 may appear as multiple different transaction cards when performing contactless data transactions, each with a unique identity, that is specific to the card frame 101 alone (with the receptacle 103 empty), or specific to the unique combinations of the card frame 101 and each different transaction card 102 inserted in the receptacle. Moreover, the transaction cards 102 also appear unique with their own respective identities when performing a data transaction without the card frame.

[0068] One or more aspects discussed herein may be embodied in computer-usable or readable data and/or computer-executable instructions, such as in one or more program modules, executed by one or more computers or other devices as described herein. Generally, program modules include routines, programs, objects, components, and data structures that perform particular tasks or implement particular abstract data types when executed by a processor in a computer or other device. The modules may be written in a source code programming language that is subsequently compiled for execution, or may be written in a scripting language such as (but not limited to) HTML or XML. The computer executable instructions may be stored on a computer readable medium such as solid-state memory, RAM, and the like. As will be appreciated by one of skill in the art, the functionality of the program modules may be combined or distributed as desired in various embodiments. In addition, the functionality may be embodied in whole or in part in firmware or hardware equivalents such as integrated circuits, field programmable gate arrays (FPGA), and the like. Particular data structures may be used to more effectively implement one or more aspects discussed herein, and such data structures are contemplated within the scope of computer executable instructions and computer-usable data described herein. Various aspects discussed herein may be embodied as a method, a computing device, a system, and/or a computer program product.

[0069] Although the present invention has been described in certain specific aspects, many additional modifications and variations would be apparent to those skilled in the art. In particular, any of the various processes described above may be performed in alternative sequences and/or in parallel (on different computing devices) in order to achieve similar

results in a manner that is more appropriate to the requirements of a specific application. It is therefore to be understood that the present invention may be practiced otherwise than specifically described. Thus, embodiments of the present disclosure should be considered in all respects as illustrative and not restrictive.

What is claimed is:

1. A card device operating as a transaction card for performing a data transaction with a terminal, the card device comprising:

- a flat sheet;
- a memory;
- an antenna integrated in the flat sheet for performing short-range wireless communication with the terminal; and
- a processor circuit integrated in the flat sheet, wherein the processor circuit is configured to:
 - receive, via the antenna, a first short-range wireless communication from the terminal indicating a data transaction;
 - retrieve from the memory a credential;
 - authenticate the card device based on the credential;
 - perform, via the antenna, a second short-range wireless communication between the card device and the terminal based on the credential; and
 - complete, via the antenna, a data transaction between the card device and the terminal, wherein the first short-range wireless communication and the second short-range wireless communication convey data to perform the data transaction.

2. The card device of claim 1, wherein the card device is made of plastic or metal.

3. The card device of claim 1, wherein the antenna is exposed along an edge of the card device.

4. The card device of claim 1, wherein the antenna is partially but not completely surrounded by the flat sheet.

5. The card device of claim 1, wherein the flat sheet is metal and the antenna is insulated from the flat sheet.

6. The card device of claim 1, wherein the card device is one of a card frame or transaction card.

7. The card device of claim 6, wherein the card device is the transaction card, the transaction card comprising a computer chip configured to execute one or more applications to perform one or more of cryptographic processing, authentication, digitally signing payment data, or cardholder verification associated with data transactions between the transaction card and the terminal.

8. The card device of claim 1, wherein the card device is a card assembly combining a transaction card with a card frame.

9. The card device of claim 8, wherein the card frame comprises:

- a receptacle integrated in the flat sheet and configured to secure the transaction card in the card frame.

10. The card device of claim 8, wherein the transaction card comprises a computer chip configured to execute one or more applications to perform one or more of cryptographic processing, authentication, digitally signing payment data, or cardholder verification associated with data transactions between the transaction card and the terminal.

11. The card device of claim 1, wherein the short-range wireless communication is NFC communication.

12. A card assembly for performing a data transaction with a terminal, the card assembly comprising:

- a metal card frame comprising:
 - a flat metal sheet;
 - a receptacle integrated in the flat metal sheet and configured to secure a transaction card in the metal card frame; a first antenna configured to wirelessly convey information between the metal card frame and a transaction card secured in the receptacle;
 - a second antenna configured to communicate with the terminal via short-range wireless communication; and
 - a processor circuit configured to perform the data transaction, via the second antenna, with the terminal based on the information.

13. The card assembly according to claim 12, further comprising the transaction card, the transaction card comprising:

- a third antenna configured to wirelessly receive and transmit the information wirelessly conveyed between the metal card frame and the transaction card by the first antenna,
- wherein the third antenna is further configured to communicate with the terminal via short range wireless communication.

14. The card assembly of claim 13, further comprising a power circuit configured to:

- convert radio frequency power received via the second antenna to direct-current power;
- supply the direct-current power to the processor circuit; and
- provide power, via the first antenna, to the transaction card.

15. The card assembly of claim 13, wherein the transaction card comprises a computer chip configured to execute one or more applications to perform one or more of cryptographic processing, authentication, digitally signing payment data, or cardholder verification associated with data transactions between the transaction card and the terminal.

16. The card assembly of claim 12, wherein the receptacle, the first antenna, and the processor circuit are integrated in the flat metal sheet.

17. The card assembly of claim 12, further comprising an electromagnetic shield configured to prevent wireless communication directly between the terminal and the transaction card when the transaction card is secured in the receptacle.

18. The card assembly of claim 12, wherein the processor circuit comprises a power circuit configured to power the transaction card via the first antenna.

19. The card assembly of claim 18, wherein the power circuit receives power via the second antenna and provides the power to the transaction card via the first antenna.

20. The card assembly of claim 12, wherein the short-range wireless communication is NFC communication.

21. A card device operating as a transaction card for performing a data transaction with a terminal, the card device comprising:

an antenna configured to perform a data transaction between the card device and a terminal via short-range wireless communication; and

a computer chip configured to:

- authenticate the data transaction with one or more credentials; and
- perform the data transaction with the terminal based on authentication of the one or more credentials.

22. The card device according to claim 21, wherein the computer chip is configured to execute one or more applications to perform one or more of cryptographic processing, authentication, digitally signing payment data, or cardholder verification associated with the data transaction between the card device and the terminal.

23. The card device of claim 21, wherein the short-range wireless communication is NFC communication.

24. A system for performing a data transaction, the system comprising:

- a server configured to provide authorized cardholders with access to account information;
- a terminal configured to read a transaction card, the terminal comprising:

- a network interface communicatively coupled to the server via a communication network to authorize a data transaction; and

- a card interface for short range wireless communication with transaction cards;

- a card device operating as a transaction card, the card device comprising:

- a memory;
- a communication interface for performing short-range wireless communication with the terminal via the card interface; and

- a processor circuit, wherein the processor circuit is configured to:

- receive, via the communication interface, a first short-range wireless communication from the terminal via the card interface indicating a data transaction;

- retrieve from the memory a credential;

- authenticate the card device based on the credential;

- perform, via the communication interface, a second short-range wireless communication between the card device and the terminal via the card interface based on the credential; and

- complete, via the communication interface and the card interface, a data transaction between the card device and the terminal, wherein the first short-range wireless communication and the second short-range wireless communication convey data to perform the data transaction.

25. The system of claim 24, wherein the first short-range wireless communication and the second short-range wireless communication are NFC communication.

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